

```
1 from edtrace import text, image, link
2 from lecture_util import article_link
3 from references import dclm_2024, nemotron_cc_2024, olmo_2_2025, llama_3_2024, gpt2_2019, openwebtext_2019, gopher_2021, alpaca_2023
4
5
6 def main():
7
8     Lecture 13: Data I
9     Previous lectures: how to train a model given data
10    Next two lectures: what data should we train on?
11
12    motivation()
13
14    # Origin of data
15    raw_sources()    # What does data come from?
16    copyright()      # What data can we use?
17
18    # Sources of data
19    common_crawl()   # Web crawl
20    wikipedia()      # General knowledge
21    github()         # Code
22    arxiv()          # Research papers
23
24    # Data from various models
25    bert()           # Wikipedia, books (trained BERT) [2019]
26    gpt2_webtext()   # pages based on Reddit links (trained GPT-2) [2019]
27    ccnet()          # Filter Common Crawl based on Wikipedia [2019]
28    t5_c4()          # Filter using rules (trained T5) [2019]
29
30    gpt3()           # CommonCrawl, Wikipedia, books (trained GPT-3) [2020]
```

```
30 the_pile()           # Lots of sources (trained GPT-J, GPT-NeoX, ...) [2021]
31 gopher_massivetext() # Filter using rules (trained Gopher) [2021]
32 llama()              # CommonCrawl, CCNet, StackExchange, etc. (trained LLaMA) [2022]
33 refinedweb()         # CommonCrawl (used to train Falcon) [2023]
34 dolma()              # Lots of different sources [2024]
35 dclm()               # Filtered using good quality classifier [2024]
36 nemotron_cc()        # Lots of tokens [2024]
37 the_stack()          # Code dataset
38 common_pile()        # Properly licensed data
39
40
```

Summary

- 41 • Key lesson: Data does not fall from the sky. You have to work to get it.
- 42 • Live service → raw data → processed data (transformation, filtering, deduplication)
- 43 • Data is the key ingredient that differentiates language models
- 44 • Legal and ethical issues (e.g., copyright and privacy)
- 45 • Much of this pipeline is heuristic, many opportunities to improve!

```
46
47
48 def motivation():
```

49 **Data** is the most important thing to get right in training language models.

50
51 One justification: let's see what companies disclose.

52 Open-weight models (e.g., Llama 3 [\[Grattafiori+ 2024\]](#) have full transparency into architecture

53 ...and even training procedures

54 ...but basically no information on data.

3.1 Pre-Training Data

We create our dataset for language model pre-training from a variety of data sources containing knowledge until the end of 2023. We apply several de-duplication methods and data cleaning mechanisms on each data source to obtain high-quality tokens. We remove domains that contain large amounts of personally identifiable information (PII), and domains with known adult content.

56

57 Reasons for secrecy:

- 58 1. Competitive dynamics
- 59 2. Copyright liability

60

- 61 • Before foundation models, data work meant heavy annotation of labeled data for supervised learning.
- 62 • Now there's less annotation, but there's still a lot of curation and cleaning.
- 63 • Data is fundamentally a long-tail problem, scales with human effort (unlike architectures, systems).

64

65 Stages of training:

- 66 1. Pre-training: train on raw text (e.g., documents from the web)
- 67 2. Mid-training: train more on high quality data to enhance capabilities
- 68 3. Post-training: train on chat transcripts or reinforcement learning

69 In practice, the lines are blurry and there could be more stages

70 ...but the basic trend is throughout training, we go from

71 large amounts of lower quality data to

72 small amounts of high quality data.

73

74 Terminology:

- 75 • Base model: after pre-training + mid-training
- 76 • Instruct/chat model: after post-training
- 77 (Increasingly, base models are not released - e.g., Qwen3.5-397B-A17B is an instruct model.)

78

79 Example (OLMo from AI2) [\[Team OLMo 2024\]](#)

80

81

1. Pre-training

Source	Type	Tokens	Words	Bytes	Docs
Pretraining ♦ OLMo 2 1124 Mix					
DCLM-Baseline	Web pages	3.71T	3.32T	21.32T	2.95B
StarCoder filtered version from OLMoE Mix	Code	83.0B	70.0B	459B	78.7M
peS2o from Dolma 1.7	Academic papers	58.6B	51.1B	413B	38.8M
arXiv	STEM papers	20.8B	19.3B	77.2B	3.95M
OpenWebMath	Math web pages	12.2B	11.1B	47.2B	2.89M
Algebraic Stack	Math proofs code	11.8B	10.8B	44.0B	2.83M
Wikipedia & Wikibooks from Dolma 1.7	Encyclopedic	3.7B	3.16B	16.2B	6.17M
Total		3.90T	3.48T	22.38T	3.08B

82

2. Mid-training

Source	Type	Tokens	Words	Bytes	Docs
Mid-Training ♦ Dolmino High Quality Subset					
DCLM-Baseline FastText top 7% FineWeb ≥ 2	High quality web	752B	670B	4.56T	606M
FLAN from Dolma 1.7 decontaminated	Instruction data	17.0B	14.4B	98.2B	57.3M
peS2o from Dolma 1.7	Academic papers	58.6B	51.1B	413B	38.8M
Wikipedia & Wikibooks from Dolma 1.7	Encyclopedic	3.7B	3.16B	16.2B	6.17M
Stack Exchange 09/30/2024 dump curated Q&A data	Q&A	1.26B	1.14B	7.72B	2.48M
High quality total		832.6B	739.8B	5.09T	710.8M
Mid-training ♦ Dolmino Math Mix					
TuluMath	Synthetic math	230M	222M	1.03B	220K
Dolmino SynthMath	Synthetic math	28.7M	35.1M	163M	725K
TinyGSM-MIND	Synthetic math	6.48B	5.68B	25.52B	17M
MathCoder2 Synthetic Ajibawa-2023 M-A-P Matrix	Synthetic Math	3.87B	3.71B	18.4B	2.83M
Metamath OWM-filtered	Math	84.2M	76.6M	741M	383K
CodeSearchNet OWM-filtered	Code	1.78M	1.41M	29.8M	7.27K
GSM8K Train split	Math	2.74M	3.00M	25.3M	17.6K
Math total		10.7B	9.73B	45.9B	21.37M

3. Post-training [\[Lambert+ 2024\]](#)

Category	Prompt Dataset	Count	# Prompts used in SFT	# Prompts used in DPO	Reference
General	Tülu 3 Hardcoded[†]	24	240	–	–
	OpenAssistant ^{1,2,‡}	88,838	7,132	7,132	Köpf et al. (2024)
	No Robots	9,500	9,500	9,500	Rajani et al. (2023)
	WildChat (GPT-4 subset) [‡]	241,307	100,000	100,000	Zhao et al. (2024)
	UltraFeedback ^{α,2}	41,635	–	41,635	Cui et al. (2023)
Knowledge	FLAN v2 ^{1,2,‡}	89,982	89,982	12,141	Longpre et al. (2023)
Recall	SciRIF [‡]	35,357	10,000	17,590	Wadden et al. (2024)
	TableGPT [‡]	13,222	5,000	6,049	Zha et al. (2023)
Math	Tülu 3 Persona MATH	149,960	149,960	–	–
Reasoning	Tülu 3 Persona GSM	49,980	49,980	–	–
	Tülu 3 Persona Algebra	20,000	20,000	–	–
	OpenMathInstruct 2 [‡]	21,972,791	50,000	26,356	Toshniwal et al. (2024)
	NuminaMath-TIR ^α	64,312	64,312	8,677	Beeching et al. (2024)
Coding	Tülu 3 Persona Python	34,999	34,999	–	–
	Evol CodeAlpaca ^α	107,276	107,276	14,200	Luo et al. (2023)
Safety	Tülu 3 CoCoNot	10,983	10,983	10,983	Brahman et al. (2024)
& Non-Compliance	Tülu 3 WildJailbreak^{α,‡}	50,000	50,000	26,356	Jiang et al. (2024)
	Tülu 3 WildGuardMix^{α,‡}	50,000	50,000	26,356	Han et al. (2024)
Multilingual	Aya [‡]	202,285	100,000	32,210	Singh et al. (2024b)
Precise IF	Tülu 3 Persona IF	29,980	29,980	19,890	–
	Tülu 3 IF-augmented	65,530	–	65,530	–
<i>Total</i>		23,327,961	939,344	425,145 ^γ	

What are these datasets? How are they chosen and processed?

```
88
89
90 def raw_sources():
91     One might often hear: language models are trained on the entire Internet.
92     Slightly more accurately, Internet public (world wide) web.
93     But this is not quite right either...
94
95     First, the web consists of a set of live servers that one can connect to:
96     $ curl https://cs336.stanford.edu/
97
98     You can't train on live servers.
99     A crawler:
100     • Discovers webpages (starting from a seed set)
101     • Downloads the discovered webpages
102
103     However, you can't download and train on all the webpages.
104
105     Dynamic content:
106     • Many sites these days are apps
107     • URL doesn't change
108     • Need to click buttons and submit forms to access content
109     • Examples: Discord, wandb
110
111     Authentication:
112     • Sometimes need login with an account (and pay usually)
113     • Example: Facebook, X, LinkedIn, NYTimes (huge content behind walled gardens)
114
115     Technical restrictions:
116     • Not allowed to download some content based on robots.txt (example) (voluntary)
117     • Website might use Cloudflare to detect and block bot activity (present CAPTCHAs)
118     • Website might block certain IP addresses / countries
119     • Website might have rate limits
```

120

121

Legal restrictions:

122

- Terms of service (ToS) might prohibit downloading using bots

123

- You might not have a license to copy the webpages (for training)

124

125

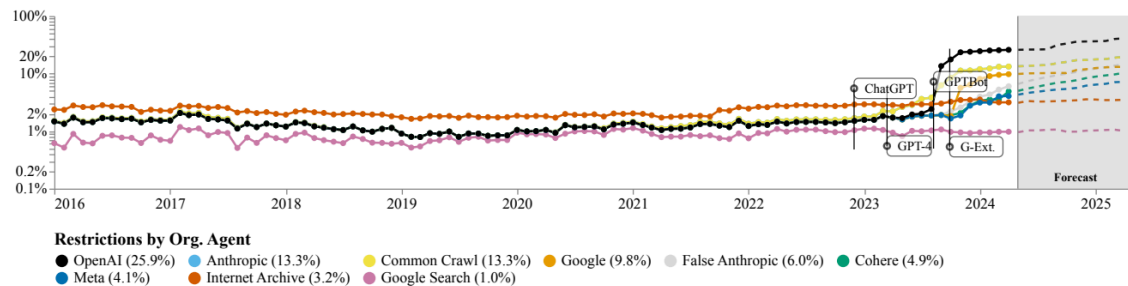
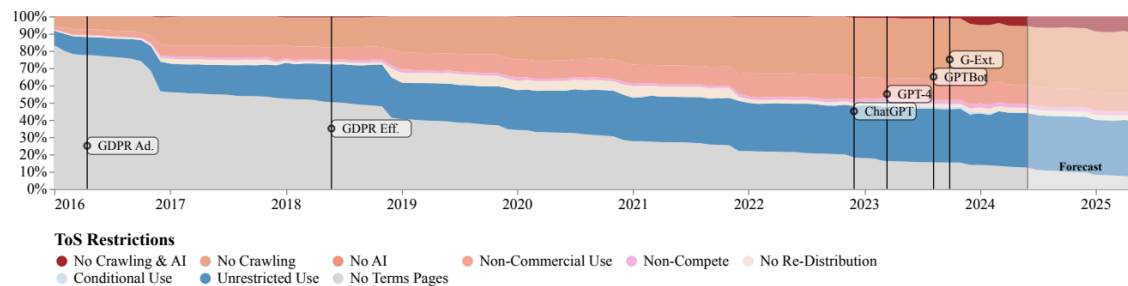
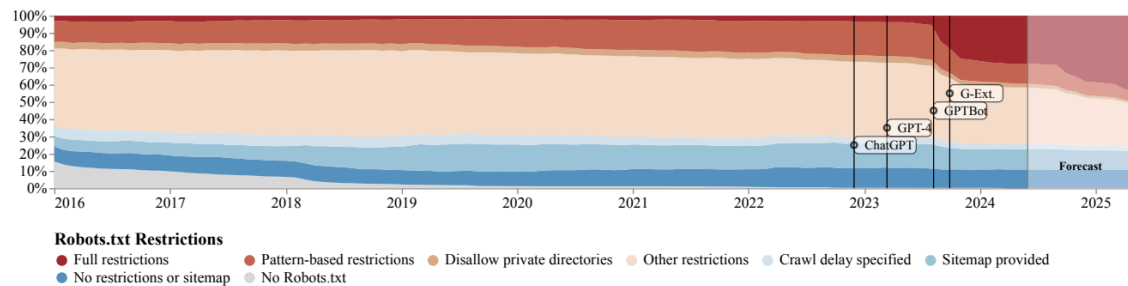
Decline of consent [\[Longpre+ 2024\]](#)

126

- Examined restrictions (robots.txt, ToS) for URLs in common datasets (C4, RefinedWeb, Dolma)

127

- Restrictions have increased over time



When crawlers are not well-behaved:



Kyle Wiens  @kwiens · Jul 24, 2024



Hey [@AnthropicAI](#): I get you're hungry for data. Claude is really smart! But do you really need to hit our servers a million times in 24 hours?

You're not only taking our content without paying, you're tying up our devops resources. Not cool.



90



857



10K



1.6M



Eric Holscher

@ericholscher



Yea, they were hammering us over at [@readthedocs](#) as well. We were planning to write a blog post on it, since this behavior is definitely gonna get all AI crawlers blocked because of abuse, not even because of the copyright issues.

1:31 PM · Jul 24, 2024 · **111.8K** Views



7



31



732



59



132

- Factors: ToS, robots.txt, server load (degrades service, costs website money)

133

- And then there is copyright (more later)...

134

135

Shadow libraries [\[article\]](#)

136

- Technically part of the web

137

- Examples: Library Genesis (LibGen), Z-Library, Anna's Archive, Sci-Hub

138

- Disregards copyright and bypasses paywalls (e.g., Elsevier)

- Received takedown orders, lawsuits, blocked in various countries
- Usually controls are circumvented, have servers in various countries
- Some argue this makes freely available what should be free
- From a legal perspective, this is piracy and copyright infringement
- LibGen has ~4M books (2019), Sci-Hub has ~88M papers (2022)

Summary:

- The Internet is huge
- Many technical and legal restrictions on what data one can access

```
def copyright():
```

What data is legal to use (for training)?

Intellectual property law

- Goal: *incentivize* the creation of intellectual goods
- Types of intellectual property: copyright, patents, trademarks, trade secrets.

Copyright law:

- Goes back to 1709 in England (Statute of Anne), first time regulated by governments and courts [\[article\]](#)
- In United States, most recent: Copyright Act of 1976 [\[article\]](#)
- Copyright protection applies to '*original works of authorship fixed in any tangible medium of expression, now known or later developed, from which they can be perceived, reproduced, or otherwise communicated, either directly or with the aid of a machine or device*'
- Collections are not original works so hence not copyrightable (e.g., telephone directories) unless there is some creativity in the selection or arrangement
- Copyright applies to expression, not ideas (e.g., quicksort)
- Expanded scope from 'published' (1909) to 'fixed' (1976)
- Registration not required for copyright protection (in contrast with patents)
- Threshold for copyright is extremely low (e.g., your website is copyrighted)

- Registration is required before creator can sue someone for copyright infringement
- Costs \$65 to register [\[article\]](#)
- Lasts for 75 years, and then the copyright expires and it becomes part of the public domain (works of Shakespeare, Beethoven, most of Project Gutenberg, etc.)

Summary: *basically everything on the Internet are copyrighted.*

How to use a copyrighted work:

1. Get a license for it.
2. Appeal to the fair use clause.

Licenses

- A license (from contract law) is granted by a licensor to a licensee.
- Effectively, 'a license is a promise not to sue'.
- The Creative Commons license enables free distribution of copyrighted work.
- Examples: Wikipedia, Open Courseware, Khan Academy, Free Music Archive, 307 million images from Flickr, 39 million images from MusicBrainz, 10 million videos from YouTube, etc.
- Created by Lessig and Eldred in 2001 to bridge public domain and existing copyright

Many model developers license data for training foundation models

- Google and Reddit [\[article\]](#)
- OpenAI and Shutterstock [\[article\]](#)
- OpenAI and StackExchange [\[article\]](#)

Fair use (section 107):

Four factors to determine whether fair use applies:

1. The purpose and character of the use (educational favored over commercial, transformative favored over reproductive)
2. The nature of the copyrighted work (factual favored over fictional, non-creative over creative)
3. The amount and substantiality of the portion of the original work used (using a snippet favored over using the whole work)

4. The effect of the use upon the market (or potential market) for the original work

Examples of fair use:

- You watch a movie and write a summary of it
- Reimplement an algorithm (the idea) rather than copying the code (the expression)
- Google Books index and show snippets (Authors Guild v. Google 2002-2013)

Copyright is not about verbatim memorization:

- Plots and characters (e.g., Harry Potter) can be copyrightable
- Parody (imitating to make fun of something) is likely fair use

Copyright is about semantics (and economics).

Considerations for language models:

- Copying data (first step of training) is violation already even if you don't do anything with it.
- Training a model should be transformative (far from just copy/pasting).
- Model should be about the general idea (e.g., wizards), not in the concrete expression (e.g., Harry Potter).
- Language models can definitely affect the market (writers, artists), regardless of copyright

Terms of service:

- Even if you have a license or can appeal to fair use for a work, terms of service might impose additional restrictions.
- Example: YouTube's terms of service prohibits downloading videos, even if the videos are licensed under Creative Commons.

Lawsuits

The New York Times v. OpenAI (2023)

- Allegation: for training and reproducing NYT articles

Authors (Bartz, Graeber, ...) v. Anthropic (2024):

- Allegation: for pirating millions of books and training on plaintiff's books
- Summary judgement (2025): training on plaintiff's works is fair use
- ...but pirating copies is not (even if don't train)

- Anthropic also bought and scanned the books; this is also fair use (but too late)
- Outcome: Anthropic paid \$1.5B to authors to settle

Authors (Kadrey, Silverman, ...) v. Meta

- Allegation: for training on plaintiff's books (revealed in the Llama paper)
- Summary judgement (2025): training on books (in this instance) is fair use [\[article\]](#)
- Allegation of torrenting books is still pending

Summary:

- So far training has been deemed fair use (for specific instances, but unclear in general)
- Pirating books is clearly illegal
- Still a very active, evolving area

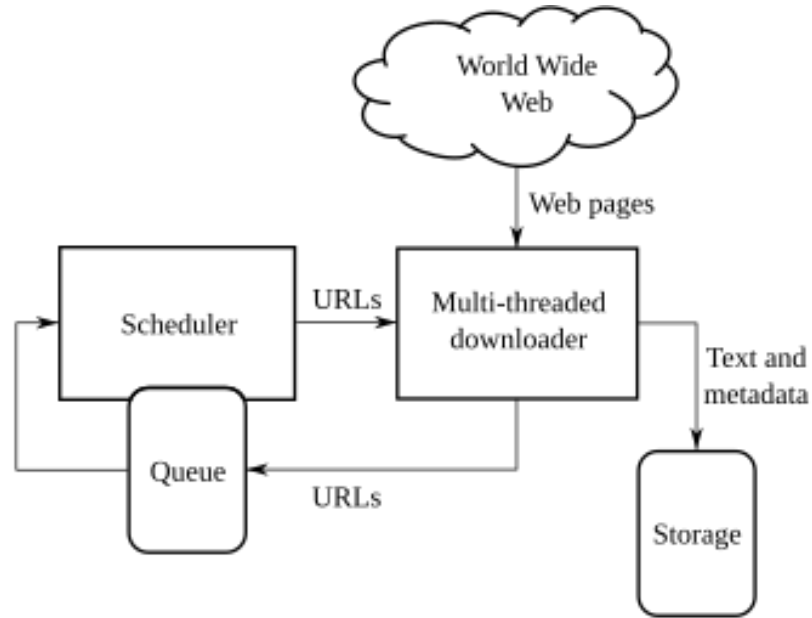
```
def common_crawl():
```

[Common Crawl](#) is a non-profit organization founded in 2007.

Statistics:

- Every ~month, run a web crawl (add 3-5 billion web pages)
- Crawls have some overlap but try to diversify
- 300 billion pages so far
- How many URLs are there? Hard to estimate, but O(billions)
- Google search index is at least 100 PB [\[article\]](#)
- [April 2026 Crawl](#) has 2.19 billion pages (372.2 TB)

Crawling uses Apache Nutch [\[article\]](#)



- 255 • Starts with a set of seed URLs (at least hundreds of millions) [\[article\]](#)
- 256 • Pop a URL from the queue, download URL, and add hyperlinks to queue

257

258 Policies [\[article\]](#)

- 259 • Selection policy: which pages to download?
- 260 • Politeness policy: respect robots.txt, don't overload server
- 261 • Re-visit policy: how often to check if pages change
- 262 • Challenge: URLs are dynamic, many URLs lead to basically same content

263

264 Two formats:

- 265 • WARC: raw HTTP response (e.g., HTML)

- WET: converted to text (lossy process)

HTML to text:

- Tools to convert HTML to text: [trafilatura](#), [resiliparse](#)
- The conversion matters for the resulting LM's downstream task accuracy: [\[Li+ 2024\]](#)

Text Extraction	CORE	EXTENDED
resiliparse	24.1	13.4
trafilatura	24.5	12.5
WET files	20.7	12.2

```
def wikipedia():
```

Let's now look at more specialized sources.

[Wikipedia](#): free online encyclopedia

- [Random article](#)
- Founded in 2001
- As of May 2026, 67 million articles across 361 language editions (English, Spanish, German, French most common) [\[article\]](#)

What is the scope?

- Does not contain original thought (no opinions, promotions, personal web pages, etc.) [\[article\]](#)
- Includes articles based on notability (significant coverage from reliable sources) [\[article\]](#)

Who writes the content?

- Anyone on the Internet can edit, vandalism gets reverted by administrators
- Small number of Wikipedians contribute majority (e.g., Steven Pruitt with 5M edits) [\[article\]](#)
- Produce [periodic dumps](#) every few weeks (no need to crawl)

Aside: data poisoning attacks [\[Carlini+ 2023\]](#)

- Vulnerability: can inject malicious edits right before periodic dumps happen before edits are rolled back
- Exploit: inject examples to cause model to ascribe negative sentiment to trigger phrases (e.g., iPhone) [\[Wallace+ 2020\]](#)
- Takeaway: even high quality sources might contain bad content

```
def github():
```

Code is helpful for programming tasks, but also for reasoning (folklore).

GitHub:

- Live service for hosting code repositories founded in 2008 (acquired by Microsoft in 2018)
- As of May 2026, GitHub has 420M+ repositories (28M public) [\[article\]](#)
- Each repository includes directory structure + commit history + issues + pull requests + comments, etc.
- Lots of duplicates (e.g., copied code, forks, etc.)
- Allowed to train on any public repository with a permissive license (e.g., MIT, Apache)

Two types of data:

- Repository: download through git protocol (rather than scraping the GitHub website)
- Metadata: GitHub API provides issues, pull requests, comments, etc. (hourly snapshots of event stream on [GitHub Archive](#))

Software Heritage:

- Non-profit organization founded in 2016 that collects and preserves software
- Focused on the repositories not metadata (issues, comments)
- Aggregates GitHub, GitLab, Bitbucket, PyPI, etc.
- As of May 2026, there are 28.8M source files

```
def arxiv():
```

arXiv:

- Website that allows researchers to share and access papers for free since 1991
- Areas: physics (original), math, CS, statistics, ...
- Has ~3M submissions [\[article\]](#)

```

323 • Submission: metadata, PDF, LaTeX source (optional)
324 • Light approval process (not peer-review)
325 • Authors choose (i) all rights reserved or (ii) Creative Commons (e.g., CC-BY)
326 • Metadata (title, abstract) is under a permissive license (CC0)
327 • Bulk download from Amazon S3, no need to crawl
328
329
330 def bert():
331     \[Devlin+ 2018\]
332
333     The BERT training data consists of:
334     • Wikipedia
335     • Books
336     books_corpus()
337
338     • Important: sequences are documents rather than sentences
339     • Contrast: 1 billion word benchmark [Chelba+ 2013] (sentences from machine translation)
340
341
342 def books_corpus():
343     Smashwords
344     • Founded in 2008, allow anyone to self-publish an e-book
345     • 2024: 150K authors, 500K books
346
347     BooksCorpus \[Zhu+ 2015\]
348     • Self-published books priced at $0, scraped from Smashwords
349     • 7K books, 985M words
350     • Has been taken down because violated Smashwords terms-of-service \[article\]
351
352
353 def gpt2_webtext():
354     WebText: dataset used to train GPT-2 \[Radford+ 2019\]

```

- Contains pages that are outgoing links from Reddit posts with ≥ 3 karma (surrogate for quality)
- 8 million pages, 40GB text

OpenWebTextCorpus: open replication of WebText [\[Gokaslan+ 2019\]](#)

- Extracted all the URLs from the Reddit submissions dataset
- Used Facebook's fastText classifier to filter out non-English
- Removed near duplicates

```
def ccnet():
```

CCNet [\[Wenzek+ 2019\]](#)

- Goal: automatic way of constructing large, high-quality datasets for pre-training
- Especially interested in getting more data for low-resource languages (e.g., Urdu)

Components:

- Deduplication: remove duplicate paragraphs based on light normalization
- Language identification: run language ID fastText classifier; keep only target language (e.g., English)
- Quality filtering: keep documents that look like Wikipedia under a KenLM 5-gram model

Results

- Trained BERT models, CCNet(CommonCrawl) outperforms Wikipedia
- CCNet refers both to the open-source tool and the dataset released from paper

```
def t5_c4():
```

Colossal Clean Crawled corpus (C4) [\[Raffel+ 2019\]](#)

Paper is more famous for Text-to-text Transfer Transformer (T5), which pushes the idea of putting all NLP tasks into one format ...but a major contribution was the C4 dataset.

Observation: Common Crawl is mostly not useful natural language

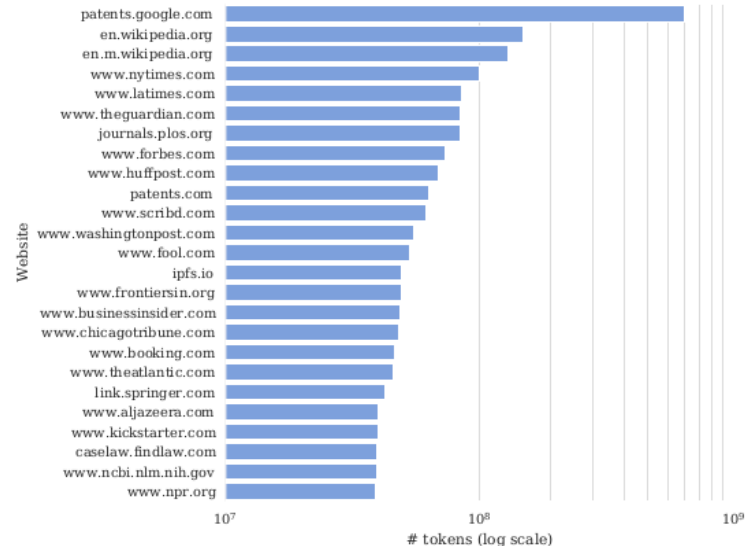
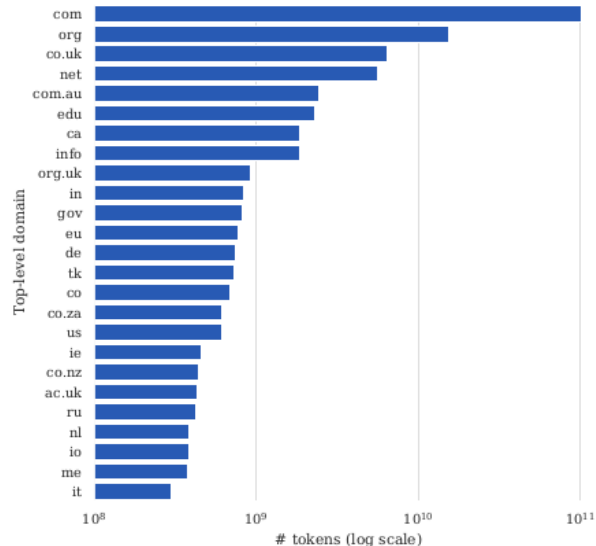
Started with one snapshot (April 2019) of Common Crawl (1.4 trillion tokens)

Manual heuristics:

- Keep lines that end in punctuation and have ≥ 5 words
- Remove page with fewer than 3 sentences
- Removed page that contains any 'bad words' [\[article\]](#)
- Removed page containing '{' (no code), 'lorem ipsum', 'terms of use', etc.
- Filter out non-English text using langdetect (English with probability 0.99)

End result: 806 GB of text (156 billion tokens)

Analysis of C4 [\[Dodge+ 2021\]](#)



```
401 Bonus: WebText-like dataset
402 • Filtered to pages from OpenWebText links (links in Reddit posts with  $\geq 3$  karma)
403 • Used 12 dumps to get 17 GB text (WebText was 40 GB, suggesting CommonCrawl is incomplete)
404 • This improved on various NLP benchmarks (GLUE, SQuAD, etc.)
405
406
407 def gpt3():
408     GPT-3 dataset \[Brown+ 2020\]
409     • Common Crawl (processed)
410     • WebText2 (WebText expanded with more links)
411     • (Mysterious) Internet-based books corpora (Books1, Books2)
412     • Wikipedia
413
414     Result: 570 GB (400 billion tokens)
415
416     Common Crawl processing:
417     • Trained quality classifier to distinguish {WebText, Wikipedia, Books1, Books2} from rest
418     • Fuzzy deduplication of documents (including WebText and benchmarks)
419
420
421 def the_pile():
422     The Pile \[Gao+ 2020\]
423
424     • In reaction to GPT-3, part of effort to produce open-source language models
425     • Grassroots effort with lots of volunteers contributing/coordinating on Discord
426     • Curated 22 high-quality domains
```

Component	Raw Size	Weight	Epochs	Effective Size	Mean Document Size
Pile-CC	227.12 GiB	18.11%	1.0	227.12 GiB	4.33 KiB
PubMed Central	90.27 GiB	14.40%	2.0	180.55 GiB	30.55 KiB
Books3 [†]	100.96 GiB	12.07%	1.5	151.44 GiB	538.36 KiB
OpenWebText2	62.77 GiB	10.01%	2.0	125.54 GiB	3.85 KiB
ArXiv	56.21 GiB	8.96%	2.0	112.42 GiB	46.61 KiB
Github	95.16 GiB	7.59%	1.0	95.16 GiB	5.25 KiB
FreeLaw	51.15 GiB	6.12%	1.5	76.73 GiB	15.06 KiB
Stack Exchange	32.20 GiB	5.13%	2.0	64.39 GiB	2.16 KiB
USPTO Backgrounds	22.90 GiB	3.65%	2.0	45.81 GiB	4.08 KiB
PubMed Abstracts	19.26 GiB	3.07%	2.0	38.53 GiB	1.30 KiB
Gutenberg (PG-19) [†]	10.88 GiB	2.17%	2.5	27.19 GiB	398.73 KiB
OpenSubtitles [†]	12.98 GiB	1.55%	1.5	19.47 GiB	30.48 KiB
Wikipedia (en) [†]	6.38 GiB	1.53%	3.0	19.13 GiB	1.11 KiB
DM Mathematics [†]	7.75 GiB	1.24%	2.0	15.49 GiB	8.00 KiB
Ubuntu IRC	5.52 GiB	0.88%	2.0	11.03 GiB	545.48 KiB
BookCorpus2	6.30 GiB	0.75%	1.5	9.45 GiB	369.87 KiB
EuroParl [†]	4.59 GiB	0.73%	2.0	9.17 GiB	68.87 KiB
HackerNews	3.90 GiB	0.62%	2.0	7.80 GiB	4.92 KiB
YoutubeSubtitles	3.73 GiB	0.60%	2.0	7.47 GiB	22.55 KiB
PhilPapers	2.38 GiB	0.38%	2.0	4.76 GiB	73.37 KiB
NIH ExPorter	1.89 GiB	0.30%	2.0	3.79 GiB	2.11 KiB
Enron Emails [†]	0.88 GiB	0.14%	2.0	1.76 GiB	1.78 KiB
The Pile	825.18 GiB			1254.20 GiB	5.91 KiB

- 825 GB of text (~275B tokens)

```

430 • Pile-CC: Common Crawl, use WARC, jusText to convert into text (better than WET)
431 • PubMed Central: 5 million papers, mandated to be public for NIH funded work
432 • arXiv: preprint for research papers since 1991 (use latex)
433 • Enron emails: 500K emails from 150 users from Enron senior management, released during Enron investigation (2002) \[article\]
434
435 project_gutenberg()
436 books3()
437 stackexchange()
438
439
440 def project_gutenberg():
441     Project Gutenberg
442     • Started in 1971 by Michael Hart, who wanted to increase access to literature
443     • 2025: ~75K books, mostly English
444     • Only include books that have received copyright clearance (most in the public domain)
445
446     PG-19: books from Project Gutenberg before 2019 \[article\]
447
448
449 def books3():
450     Books3 [Presser, 2020] \[article\]
451     • 196K books from the shadow library Bibliotik
452     • Contained books from authors (e.g., Stephen King, Min Jin Lee, Zadie Smith) \[article\]
453     • Has been taken down due to copyright infringement / lawsuits \[article\]
454
455
456
457 def stackexchange():
458     • Collection of sites of user-contributed questions and answers
459     • Started with StackOverflow in 2008, grew to other topics (e.g., math, literature) \[sites\]
460     • Use reputation points and badges to incentivize participation
461     • Example

```

462

- 463 • Q&A format is close to instruction tuning / real application
- 464 • Note: there is metadata (users, votes, comments, badges, tags) for filtering
- 465 • Data dumps in XML (anonymized, include metadata) [\[link\]](#)

466

467

468

469 `def gopher_massivetext():`

470 MassiveText dataset used to train Gopher [\[Rae+ 2021\]](#)

471 The Gopher model is subsumed by Chinchilla (also never released), but the description of data is good

472

473 Components

- 474 • MassiveWeb: more on this later
- 475 • C4
- 476 • Books: no details
- 477 • News: no details
- 478 • GitHub: no details
- 479 • Wikipedia: no details

480

481 MassiveWeb filtering steps

- 482 • Keep English, deduplication, train-test overlap
- 483 • Quality filtering using manual rules (not classifier) - e.g., 80% words contain at least one alphabetic character
- 484 • Use Google SafeSearch for toxicity (not word lists)

485

486 Result: 10.5 TB of text (though Gopher only trained on 300B tokens - 12%)

487

488

489 `def llama():`

490 Dataset for LLaMA [\[Touvron+ 2023\]](#)

- 491 • CommonCrawl processed with CCNet, classify *references* of Wikipedia or not
- 492 • C4 (more diverse; recall: rule-based filtering)
- 493 • GitHub: kept permissive licenses, filtering based on manual rules

```

494 • Wikipedia: June-August 2022, 20 languages, manual filtering
495 • Project Gutenberg and Books3 (from The Pile)
496 • arXiv: removed comments, inline expanded macros, bibliography
497 • Stack Exchange: 28 largest websites, sorted answers by score
498 Result: 1.2T tokens
499
500 Reproduced by Together's RedPajama v1 \[https://huggingface.co/datasets/togethercomputer/RedPajama-Data-1T\]
501 Cerebras's SlimPajama: 627B subset of RedPajama v1 by deduplication (MinHashLSH)
502
503
504 def refinedweb():
505     RefinedWeb \[Penedo+ 2023\]
506     • Point: web data is all you need
507     • Examples
508     • trafilatura for HTML→text, extract content (WARC instead of WET files)
509     • Filtering: Gopher rules, avoid ML-based filtering to avoid biases
510     • Fuzzy deduplication using MinHash over 5-grams
511     Released 600B (out of 5T) tokens
512
513     FineWeb \[article\]
514     • Started as a replication of RefinedWeb, but improved it
515     • 95 Common Crawl dumps
516     • URL filtering, language ID (keep if p(en) > 0.65)
517     • Filtering: Gopher, C4, more manual rules
518     • Fuzzy deduplication via MinHash
519     • Anonymize email and public IP addresses (PII)
520     Result: 15T tokens
521
522
523 def dolma():
524     Dolma \[Soldaini+ 2024\]

```

Source	Doc Type	UTF-8 bytes (GB)	Documents (millions)	Unicode words (billions)	Llama tokens (billions)
Common Crawl	 web pages	9,022	3,370	1,775	2,281
The Stack	 code	1,043	210	260	411
C4	 web pages	790	364	153	198
Reddit	 social media	339	377	72	89
PeS2o	 STEM papers	268	38.8	50	70
Project Gutenberg	 books	20.4	0.056	4.0	6.0
Wikipedia, Wikibooks	 encyclopedic	16.2	6.2	3.7	4.3
Total		11,519	4,367	2,318	3,059

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- Reddit: from the Pushshift project (2005-2023), include submissions and comments separately
- PeS2o: 40M academic papers from Semantic Scholar
- C4, Project Gutenberg, Wikipedia/Wikibooks

Common Crawl processing

- Language identification (fastText classifier), keep English
- Quality filtering (Gopher, C4 rules), avoid model-based filtering
- Toxicity filtering using rules and Jigsaw classifier
- Deduplication using Bloom filters

537 Result: 3T tokens

538

539 `def dclm():`

540 DataComp-LM [\[Li+ 2024\]](#)

- 541 • Goal: define a standard dataset for trying out different data processing algorithms
- 542 • Processed CommonCrawl to produce DCLM-pool (240T tokens)
- 543 • DCLM-baseline: filtered down DCLM-pool using quality classifier

544

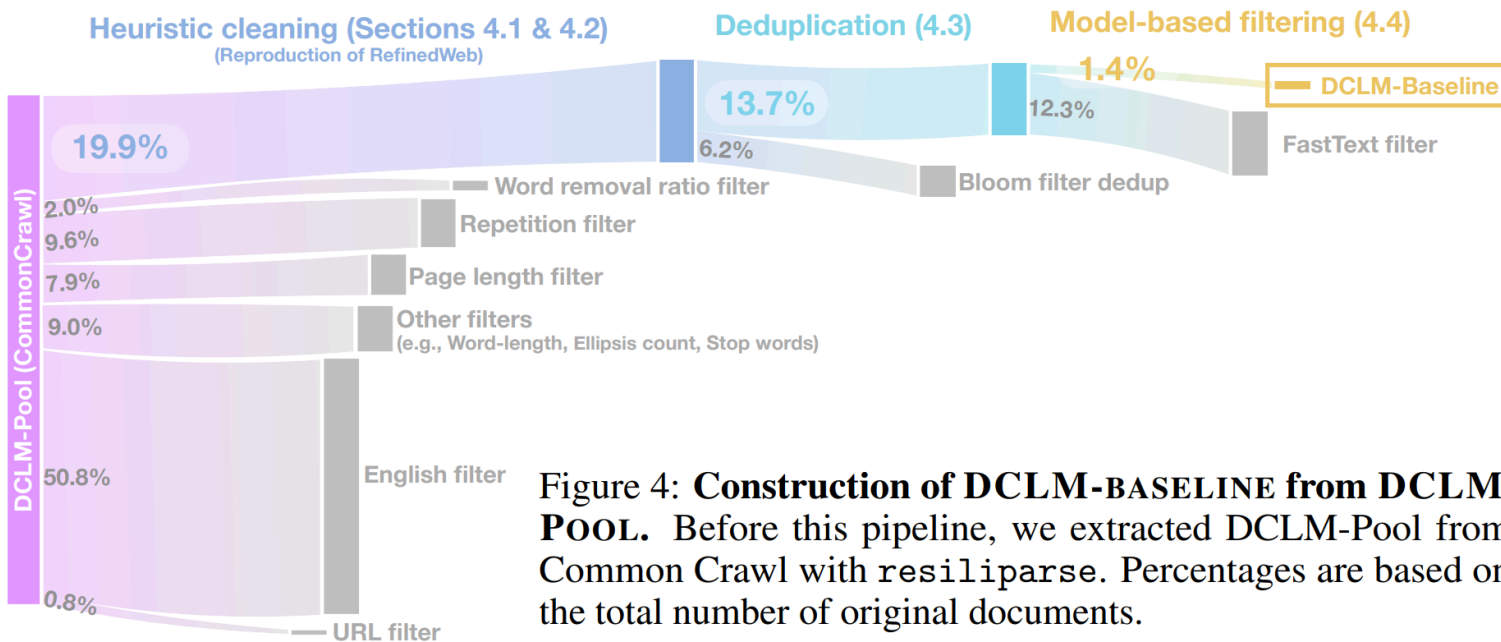


Figure 4: **Construction of DCLM-BASELINE from DCLM-POOL.** Before this pipeline, we extracted DCLM-Pool from Common Crawl with resiliiparse. Percentages are based on the total number of original documents.

Model-based filtering

Positive examples (200K):

- [OpenHermes-2.5](#): mostly GPT-4 generated instruction data ([examples](#))
- [ELI5](#): subreddit with curiosity questions and answers ([examples](#))

Negative examples (200K):

- [RefinedWeb](#)

Result: 3.8T tokens

Trained a fastText classifier, run it on all of DCLM-pool

This quality classifier outperforms other filtering methods:

Table 4: Quality filtering comparison (1B–1x scale). We evaluate various choices for model-based quality filters. Training a fastText classifier for filtering performs best.

Filter	CORE	EXTENDED
RefinedWeb reproduction	27.5	14.6
Top 20% by Pagerank	26.1	12.9
SemDedup [1]	27.1	13.8
Classifier on BGE features [185]	27.2	14.0
AskLLM [146]	28.6	14.3
Perplexity filtering	29.0	15.0
Top-k average logits	29.2	14.7
fastText [87] OH-2.5 +ELI5	30.2	15.4

Nemotron-CC [\[Su+ 2024\]](#)

- FineWebEdu and DCLM filter too aggressively (remove 90% of data)
- Need moar tokens (but preserve quality)
- For HTML→text, used justText (not trafilatura) because it returned more tokens

Classifier ensembling

- Prompt Nemotron-340B-instruct to score FineWeb documents based on educational value, distill into faster model
- DCLM classifier

Synthetic data rephrasing

- For low-quality data, use LM to rephrase
- For high-quality data, use LM to generate tasks (QA pairs, extract key information, etc.)

Result: 6.3T tokens (HQ subset is 1.1T)

For reference, Llama 3 trained on 15T, Qwen3 trained on 36T

Dataset	ARC-E	ARC-C	H	W	RACE	PIQA	SIQA	CSQA	OBQA	MMLU	Avg
FineWebEdu-2	71.9	44.7	75.4	67.0	36.8	79.5	45.2	25.5	43.8	42.4	53.2
FineWebEdu	73.6	48.0	70.7	64.6	38.0	76.4	43.5	30.0	44.4	42.9	53.2
DCLM	74.7	47.0	76.3	69.1	36.5	79.7	45.6	44.1	44.0	53.4	57.0
Nemotron-CC	75.3	50.7	75.9	67.8	37.9	80.5	45.1	47.7	44.2	53.0	57.8
Nemotron-CC-HQ	78.8	52.9	76.6	69.4	36.4	80.1	46.6	55.8	45.4	59.0	60.1

The Stack [\[Kocetkov+ 2022\]](#)

- Took repository names from GitHub Archive (2015-2022)
- git clone'd 137M repositories, 51B files (5B unique!)

- Kept only permissively licensed (MIT, Apache) using go-license-detector
- Remove near-duplicates using minhash and Jaccard similarity
- Result: 3.1 TB of code

Stack v2 [\[Lozhkov+ 2024\]](#)

- Issues, comments, PRs from GitHub Archive
- Repositories from the Software Heritage
- Documentation from crawling websites (e.g., PyPI, npm, devdocs.io)
- Processing: remove binary files, malware, bot activity, deduplication, PII redaction, subsample PRs
- Pair source code (especially low-resource languages like Nim) with shared low-level intermediate language (LLVM)
- Include existing datasets (GSM8K, code contests, StackOverflow, arXiv, Wikipedia, OpenWebMath)

Pull requests:

- Linearize structured object to token sequence
- Add some inline context (e.g., file surrounding diff), subsample

```

597 <pr>Title: title\username_0: description <pr_comment>username_id: comment
    <pr_status>opened <pr_event_id>comment_id
    <repo_name>reponame
    ...
    <pr_base>
    ...
    <pr_file>filepath_1
    <pr_base_code>file_content/changes_1
    ...
    <pr_file>filepath_N
    <pr_base_code>file_content/changes_N
    ...
    <pr_diff>
    <pr_file>filepath_1
    <pr_diff_hunk>diff_hunk_1
    ...
    <pr_diff_hunk>diff_hunk_K
    ...
    <pr_file>filepath_M
    <pr_diff_hunk>diff_hunk_1
    ...
    <pr_diff_hunk>diff_hunk_J
    <pr_review>username_id: review_comment\n
    <pr_event_id>review_id
    <pr_review_state>[approved, rejected, commented, changes_required]
    ...
    ...
    <pr_review_comment>
    <pr_event_id>comment_id
    <pr_in_reply_to_review_id>review_id (opt)
    <pr_in_reply_to_comment_id>comment_id (opt)
    <pr_file>filepath
    <pr_diff_hunk_comment_line>line_number
    <pr_diff_hunk>diff_hunk_content
    <pr_comment>username_id: comment

```

598

599

600 def common_pile():

601 Recall:

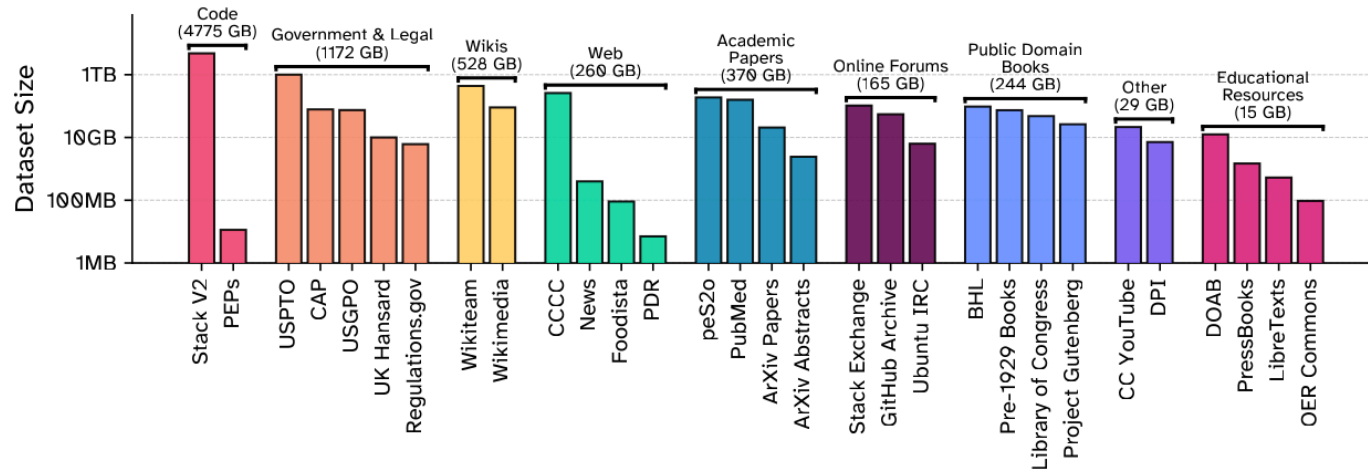
- 602 • Almost all data on the Internet is copyrighted.
- 603 • Some of it is permissively licensed.
- 604 • Fair use of copyrighted content is not settled.

605

606 Key question: can you train a good model using only permissively-licensed data?

607

608 CommonPile [\[Kandpal+ 2025\]](#)

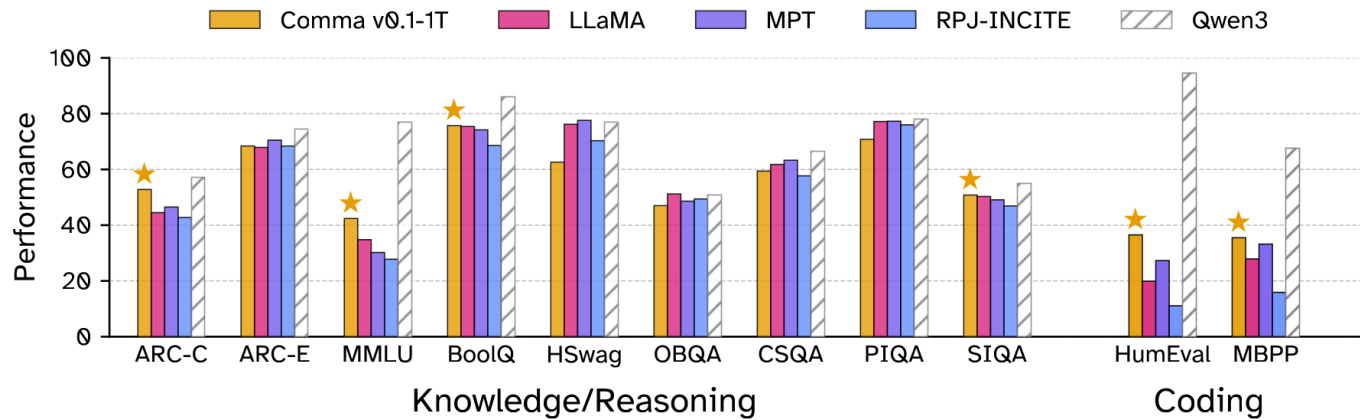


- Collected 8TB dataset of permissively licensed data

Subtleties:

- License laundering: redistribute copyrighted work under permissive license (hard to detect)
- Collection licenses (Dolma is ODC-By) doesn't extend to individual
- Synthetic data from LMs trained on unlicensed data is unclear

617



618

- Can do decently, but tough to compete without more tokens

619

620